A CECH COMPLEX OVER A COVERING CONTAINING THE WHOLE SPACE IS HOMOTOPY EQUIVALENT TO ZERO

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed. The check accent on Cech and the author's align* computation are reproduced verbatim.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot: sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The counted statement and proof contain no cross reference at all, so the assembled document defines every label it uses and no external reference stub is needed.

Allowed standard prerequisites (editorial).

- The extended Cech complex of an open covering of a topological space. Same chapter, the definition of the Cech complex with its term in degree -1 given by F(U); the statement refers to the extended complex explicitly. Supplies the complex and its indexing, including the convention U_{i_0 ... i_p} = U_{i_0} cap ... cap U_{i_p}; the homotopy is the counted content.
- Contracting homotopies of cochain complexes. Standard homological algebra: a homotopy h with dh + hd = id exhibits the complex as homotopy equivalent to zero. Supplies the shape of the verification performed in the proof.

Lemma 0.1. *Let X be a topological space. Let $\mathcal{F}$ be an abelian presheaf on X. Let $\mathcal{U} : U = \bigcup_{i \in I} U_i$ be an open covering. If $U_i = U$ for some $i \in I$, then the extended Čech complex*

$$\mathcal{F}(U) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$$

obtained by putting $\mathcal{F}(U)$ in degree -1 with differential given by the canonical map of $\mathcal{F}(U)$ into $\check{\mathcal{C}}^0(\mathcal{U}, \mathcal{F})$ is homotopy equivalent to 0.

Proof. Fix an element $i \in I$ with $U = U_i$. Observe that $U_{i_0 \ldots i_p} = U_{i_0 \ldots \hat{i}_j \ldots i_p}$ if $i_j = i$. Let us define a homotopy

$$h : \prod\nolimits_{i_0 \ldots i_{p+1}} \mathcal{F}(U_{i_0 \ldots i_{p+1}}) \longrightarrow \prod\nolimits_{i_0 \ldots i_p} \mathcal{F}(U_{i_0 \ldots i_p})$$

by the rule

$$h(s)_{i_0 \ldots i_p} = s_{i i_0 \ldots i_p}$$

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In other words, $h : \prod_{i_0} \mathcal{F}(U_{i_0}) \to \mathcal{F}(U)$ is projection onto the factor $\mathcal{F}(U_i) = \mathcal{F}(U)$ and in general the map h equals the projection onto the factors $\mathcal{F}(U_{ii_1\ldots i_{p+1}}) = \mathcal{F}(U_{i_1\ldots i_{p+1}})$. We compute

$$\begin{aligned}(dh + hd)(s)_{i_0\ldots i_p} &= \sum\nolimits_{j=0}^{p} (-1)^j h(s)_{i_0\ldots \hat{i}_j \ldots i_p} + d(s)_{ii_0\ldots i_p} \\ &= \sum\nolimits_{j=0}^{p} (-1)^j s_{ii_0\ldots \hat{i}_j \ldots i_p} + s_{i_0\ldots i_p} + \sum\nolimits_{j=0}^{p} (-1)^{j+1} s_{ii_0\ldots \hat{i}_j \ldots i_p} \\ &= s_{i_0\ldots i_p}\end{aligned}$$

This proves the identity map is homotopic to zero as desired. □